Two-Dimensional R.V.

So far we have considered only the one-dimensional R.V. i.e., we have considered such random experiments, the out come of which had only one characteristic and hence was assigned a single real value. In many situations we will be interested in recording 2 or more characteristics (numerically) of the out come of a random experiment.

For example, both voltage and current might be of interest in a certain experiment.

Definition: Let S be the sample space associated with a random experiment E. $X = X(s)$ and $Y = Y(s)$ be two (random) functions each assigns a real numbers to each out comes $s \in S$. Then (X, Y) is called two dimensional random variable.

If the possible values of (X, Y) are finite or countably infinite, (X, Y) is called a two dimensional R.V.

(X, Y) is a two dimensional discrete RV the possible values of (X, Y) may be represented as (x_i, y_j), $i = 1, 2 \ldots k \ldots m$, $j = 1, 2 \ldots n$.

If (X, Y) can assume all values in a specified region R in the XY-plane, (X, Y) is called a two dimensional continuous RV.

Probability function of (X, Y).

If (X, Y) is two dimensional discrete RV such that $P(X = x_i, Y = y_j) = P_{ij}$ then P_{ij} is called Probability mass function or simply probability function.

(i) $P_{ij} \geq 0$ for all i and j

(ii) $\sum_j \sum_i P_{ij} = 1$.

The set of triples $\{x_i, y_j, P_{ij}\}$ is called joint probability distribution of (X, Y). $i = 1, 2, \ldots m \ldots \quad j = 1, 2, \ldots k, \ldots n \ldots$

Joint Probability Density function

If (X, Y) is a two dimentional Continuous R.V such that

$$P\left\{x - \frac{dx}{2} \leq x \leq x + \frac{dx}{2};\ y - \frac{dy}{2} \leq y \leq y + \frac{dy}{2}\right\} = f(x, y)\,dx\,dy$$

then $f(x, y)$ is called the joint pdf of (X, Y). Provided $f(x, y)$ satisfies the following conditions.

(i) $f(x, y) \geq 0$ for all $(x, y) \in R$, where R is the range space

(ii) $\iint_R f(x, y)\,dx\,dy = 1$.

moreover if D is a subspace of the range space R, $\{P(X, Y) \in D\}$ is defined as

$$P\{(X, Y) \in D\} = \iint_D f(x, y)\,dx\,dy$$

In particular $P\{a \leq x \leq b,\ c \leq Y \leq d\} = \int_a^b \int_c^d f(x, y)\,dy\,dx.$

Cummulative Distribution function :

If (X, Y) is a two dimensional RV (discrete or continuous), then $F(X,Y) = P(X \le x \text{ and } Y \le y)$ is called C df of (X, Y).

In the descrete case

$$F(x,y) = \sum_j \sum_i p_{ij} \qquad y_j \le y,\ x_i \le x.$$

In the Continuous Case,

$$F(x,y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(x,y)\, dx\, dy)$$

Properties of $F(x,y)$

(i) $F(-\infty, y) = 0 = F(x, -\infty)$ and $F(\infty, \infty) = 1$

(ii) $P\{a < X < b,\ Y \le y\} = F(b,y) - F(a,y)$

(iii) $P\{X \le x,\ c < Y < d\} = F(x,d) - F(x,c)$.

(iv) $P\{a < X < b,\ c < Y < d\} = F(b,d) - f(a,d) - F(b,c) + F(a,c)$.

At any points continuity of $f(x,y)$

$$\frac{\partial^2 F}{\partial x \partial y} = f(x,y).$$

Marginal Probability distributions:

$P(X=x_i) = P(X=x_i \text{ and } Y=y_1) \text{ or } (X=x_i \text{ and } Y=y_2) \text{ or etc}$

$= P_{i1} + P_{i2} + K = \sum_j P_{ij}$

$P(X=x_i) = \sum_j P_{ij}$ is called the marginal probability function of X. It is defined for $X = x_1, x_2 \ldots$ and denoted as P_{i*}. The collection of pairs $\{x_i, P_{i*}\}$, $i = 1,2,3\ldots$ is called marginal probability distribution of X. Similarly the collection of pairs $(y_j, P_{*j}) = j = 1,2,\ldots$ is called marginal probability distribution of Y.

Where $P_{*j} = \sum_i P_{ij} = P(Y=y_j)$.

In the continuous case.

$P\{x - \frac{1}{2}dx \le X \le x + \frac{1}{2}dx, -\infty \le y \le \infty\}$

$= \int_{-\infty}^{\infty} \int_{x-\frac{1}{2}dx}^{x+\frac{1}{2}dx} f(x,y)\, dx\, dy.$

$= \left[\int_{-\infty}^{\infty} f(x,y)\, dy\right] dx$ [Since $f(x,y)$ may be treated as constant in $(x-\frac{1}{2}dx, x+\frac{1}{2}dx)$.

$= f_x(x)\, dx$ say.

Illy $f_y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$ is called the marginal density of Y.

Note: $P(a \le X \le b) = P(a \le X \le b, -\infty \le Y \le \infty)$

$$= \int_{-\infty}^{\infty} \int_{a}^{b} f(x,y)\,dx\,dy$$

$$= \int_{a}^{b} \left[\int_{-\infty}^{\infty} f(x,y)\,dy \right] dx = \int_{a}^{b} f_x(x)\,dx$$

Illy $P(c \le Y \le d) = \int_{c}^{d} f_y(y)\,dy$

Conditional Probability Distribution:

$P(X = x_i / Y = y_i) = \dfrac{P(X = x_i, Y = y_i)}{P(Y = y_i)} = \dfrac{p_{ij}}{p_{xj}}$ is called the conditional probability function of X given that $Y = y_j$

The collection of pairs $\left\{ x_i, \dfrac{p_{ij}}{p_{i*}} \right\}$, $j = 1, 2, 3, \ldots$ is called the conditional probability distribution of Y given $X = x_i$.

In the continuous case,

$$P\{x - \tfrac{1}{2}dx \le X \le x + \tfrac{1}{2}dx \mid Y = y\}$$

$$= P\left\{x - \tfrac{1}{2}dx \le X \le x + \tfrac{1}{2}dx \,/\, y - \tfrac{1}{2}dy \le y + \tfrac{1}{2}dy\right\}.$$

$$= \frac{f(x,y)\,dx\,dy}{f_y(y)\,dy} = \left\{\frac{f(x,y)}{f_y(y)}\right\} dx.$$

$\frac{f(x,y)}{f_y(y)}$ is called the conditional density of X given Y.

and is denoted by $f(x/y)$

Illy $\frac{f(x,y)}{f_x(x)}$ is called the conditional density of Y given X

and is denoted by $f(y/x)$.

Independent RVs.

If (X,Y) is two dimensional d.r.v such that

$P(X=x_i/Y=y_i) = P(X=x_i)$ i.e., $\frac{P_{ij}}{P_{*j}} = P_{i*}$

i.e., $P_{ij} = P_{i*} \times P_{*j}$ $\forall$ i,j then X and Y are called independent RV's

Illy if (X,Y) is a two dimensional C.RV such that $f(x,y) = f_x(x) \times f_y(y)$ then X and Y are said to be Independent RVs.

Prob: Three balls are drawn at random without replacement from a box containing 2 white 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn. Find the Probability distribution of (X, Y).

Sol: As there are only 2 white balls in the box, X can take the values 0, 1, and 2

$P(X=0, Y=0)$ = P(drawing 3 balls none of which is white or red).

= P(all the three balls are black).

$= {}^4C_3 / {}^9C_3 = 1/21.$

$P(X=0, Y=1)$ = P(drawing 1 red ball and 2 black balls).

$= \frac{{}^3C_1 \times {}^4C_2}{{}^9C_3} = 3/14$

$P(X=0\ Y=2) = \frac{{}^3C_2 \times {}^4C_1}{{}^9C_3} = 1/7$

$P(X=0\ Y=3) = 1/84$

$P(X=1\ Y=0) = 1/7$

$P(X=1, Y=1) = 2/7$

$P(X=1, Y=2) = 1/14$

$P(X=1, Y=3) = 0$. since only 3 balls are drawn.

$P(X=2, Y=0) = 1/2$

$P(X=2, Y=1) = 1/28$

$P(X=2, Y=2) = 0$ since only 3 balls are drawn

$P(X=2, Y=3) = 0$

The joint probability distribution (X, Y) may be represented in the form of a given table as given below

X \ Y	0	1	2	3
0	$\frac{1}{21}$	$3/14$	$\frac{1}{7}$	$1/84$
1	$1/7$	$2/7$	$1/14$	0
2	$1/21$	$1/28$	0	0

Prob : 2.

Prob: For the bivariate probability distribution of (X, Y) is given below find $P(X \le 1)$, $P(Y \le 3)$, $P(X \le 1, Y \le 3)$, $P(X \le 1 / Y \le 3)$, $P(Y \le 3 / X \le 1)$ and $P(X+Y \le 4)$.

X \ Y	1	2	3	4	5	6
0	0	0	$\frac{1}{32}$	$\frac{2}{32}$	$\frac{3}{32}$	$\frac{3}{32}$
1	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
2	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{64}$	0	$\frac{2}{64}$

Sol: $P(X \le 1) = P(X=0) + P(X=1)$

$$= \sum_{j=1}^{6} P(X=0, Y=j) + \sum_{j=1}^{6} P(X=1, Y=j)$$

$$= \left(0 + 0 + \frac{1}{32} + \frac{2}{32} + \frac{2}{32} + \frac{3}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right)$$

$$= \frac{1}{4} + \frac{5}{8} = 7/8$$

$$P(Y \le 3) = P(Y=1) + P(Y=2) + P(Y=3)$$

$$= \sum_{i=0}^{2} P(X=i, Y=1) + \sum_{i=0}^{2} P(X=i, Y=2) + \sum_{i=0}^{2} P(X=i, Y=3)$$

$$= \left(0 + \frac{1}{16} + \frac{1}{32}\right) + \left(0 + 1/16 + 1/32\right) + \left(\frac{1}{32} + \frac{1}{8} + \frac{1}{64}\right) = \frac{23}{64}.$$

$$P(X \le 1, Y \le 3) = \sum_{j=1}^{3} P(X=0, Y=j) + \sum_{j=1}^{3} P(X=1, Y=j)$$

$$= \left(0 + 0 + \frac{1}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8}\right) = \frac{9}{32}$$

$$P(X \le 1 / Y \le 3) = \frac{P(X \le 1, Y \le 3)}{P(Y \le 3)} = \frac{9/32}{23/64} = 9/28 \quad \frac{18}{23}$$

$$P(Y \le 3 / X \le 1) = \frac{P(X \le 1, Y \le 3)}{P(X \le 1)} = \frac{9/32}{7/8} = 9/28.$$

$$P(X+Y \le 4) = \sum_{j=1}^{4} P(X=0, Y=j) + \sum_{j=1}^{3} P(X=1, Y=j) + \sum_{j=1}^{2} P(X=2, Y=j)$$

$$= \frac{3}{32} + \frac{1}{4} + \frac{1}{16} = \frac{13}{32}.$$

—o—

Marginal Probability Distribution of Y: $\{j, P_{*j}\}$

$$P_{*j} = \sum_{i=0}^{2} P_{ij}$$ column sums.

$Y=j$	
1	$15/72 = P_{01} + P_{11} + P_{21}$
2	$24/72 = P_{02} + P_{12} + P_{22}$
3	$33/72 = P_{03} + P_{13} + P_{23}$.

Total = 1

Conditional distribution of X given $y=1$,
is given by $\{i, P(X=i/y=1\} = \{i, P(X=i, y=1)/P(y=1)\}$

$= \{i \cdot P_{i1}/P_{*1}\}$ $i = 0, 1, 2$.

The tabular representation given below

$X=i$	P_{i1}/P_{*1}
0	$3K/15K = 1/5$
1	$15/15K = 1/3$
2	$7K/15K = 7/15$
	Total = 1.

Prob: The joint probability mass function (X,Y) is given by $P(X,Y) = K(2x+3y)$ $x=0,1,2$; $y=1,2,3$ Find all the marginal and conditional probability distributions Also find probability distribution.

Sol: The joint probability distribution of (X,Y) is given below. The relevant probabilities have been computed by using the given Law.

X	Y		
	1	2	3
0	3K	6K	9K
1	5K	8K	11K
2	7K	10K	13K.

$$\sum_{j=1}^{3}\sum_{i=0}^{2} P(x_i y_j) = 1.$$

i.e., the sum of all the probabilities in the above table is equal to 1. i.e., $72K = 1 \Rightarrow K = 1/72$.

Marginal Probability Distribution of X : $\{i, P_{i*}\}$.

$$P_{i*} = \sum_{j=1}^{3} P_{ij}$$

$x = i^{\circ}$

0 $\quad P_{01} + P_{02} + P_{03} = 3K + 6K + 9K = \frac{18}{72}$

1 $\quad P_{11} + P_{12} + P_{13} = 5K + 8K + 11K = \frac{24}{72}$

2 $\quad P_{21} + P_{22} + P_{23} = 7K + 10K + 13K = \frac{30}{72}$.

$$\text{Total} = \frac{18}{72} + \frac{24}{72} + \frac{30}{72} = \frac{72}{72} = 1.$$

The other Conditional distributions are given below.

CPD of X given Y=2.

X=i	P_{i2}/P_{*2}
0	$\frac{6K}{24K} = \frac{1}{4}$
1	$\frac{8K}{24K} = \frac{1}{3}$
2	$\frac{10K}{24K} = \frac{5}{12}$

Total = 1.

CPD of X given Y=3

X=i	P_{i3}/P_{*3}
0	$\frac{9K}{33K} = \frac{3}{11}$
1	$\frac{11K}{33K} = \frac{1}{3}$
2	$\frac{13K}{33K} = \frac{13}{33}$

Total = 1.

CPD of Y given X=0

Y=j	P_{0j}/P_{0*}
1	$\frac{3K}{18K} = \frac{1}{6}$
2	$\frac{6K}{18K} = \frac{1}{3}$
3	$\frac{9K}{18K} = \frac{1}{2}$

Total = 1.

CPD of Y given X=1.

Y=j	P_{1j}/P_{1*}
1	$\frac{5K}{24K} = \frac{5}{24}$
2	$\frac{8K}{24K} = \frac{1}{3}$
3	$\frac{11K}{24K} = \frac{11}{24}$

Total = 1.

CPD of Y given X=2.

Y=j	P_{2j}/P_{2*}
1	$\frac{7K}{30K} = \frac{7}{30}$
2	$\frac{10K}{30K} = \frac{1}{3}$
3	$\frac{13K}{30K} = \frac{13}{30}$

Total = 1.

Probability distribution of (X+Y).

Z = X + Y	P.
1	$P_{01} = 3/72$
2	$P_{02}+P_{11} = \frac{11}{72}$
3	$P_{03}+P_{12}+P_{21} = \frac{24}{72}$
4	$P_{13}+P_{22} = \frac{21}{72}$
5	$P_{23} = 13/72$
	total = 1.

Distribution Z = X+Y

We define a new random variable

$Z = X+Y$.

Since $X = \{0,1,2\}$ $Y = \{1,2,3\}$.

the possible values of Z are

$Z_{min} = 0+1 = 1$ $Z_{max} = 2+3 = 5$

So Z values 1,2,3,4,5

Collecting the probabilities

$Z=1$ only possible $x=0$ $y=1$.

$$P(Z=1) = P(0,1) = P_{01} = \frac{3}{72}$$

$Z=2$ Possible $P_{02} + P_{11} = \frac{11}{72}$

$Z=3$. $P_{03} + P_{12} + P_{21} = 1/3$.

$x=0, 3$ $x=1, y=2$ $x=2, y=1$.

$Z=4$ $P_{13} + P_{22} = \frac{21}{72} = 7/24$

$Z=5$ $P_{23} = 13/72$

Z	$P(Z)$.
~~0~~	
1	3/72
2	11/72
3	1/3
4	7/24
5	13/72

Total = 1.

Prob: The joint Pdf of a two dimensional RV (X,Y) is given by $f(x,y) = xy^2 + x^2/8$, $0 \le x \le 2$, $0 \le y \le 1$.

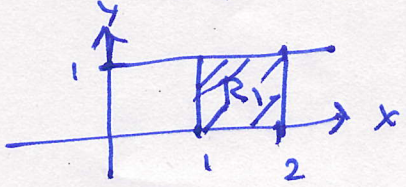
Compute $P(X>1)$, $P(Y<1/2)$, $P(X>1/Y<1/2)$ $P(Y<1/2/x>1)$, $P(X<Y)$ and $P(X+Y<1)$.

Sol: Hence the rectangle defined by $0 \le x \le 2$, $0 \le y \le 1$. is the range space R.

$R_1, R_2 \ldots R_n \ldots$ are the events spaces.

(i) $P(X>1) = \iint_{R_1 (x>1)} f(x,y)\,dx\,dy$

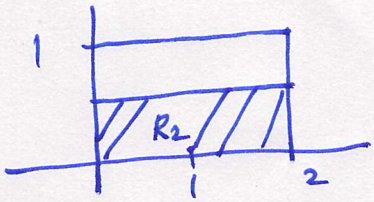
$= \int_0^1 \int_1^2 (xy^2 + x^2/8)\,dx\,dy = \frac{19}{24}$.



(ii) $P(Y<1/2) = \iint_{R_2} (xy^2 + x^2/8)\,dx\,dy$

lts. y lts 0 to 1/2
x lts 0 to 2

$= \int_0^{1/2} \int_0^2 (xy^2 + x^2/8)\,dx\,dy = 1/4$

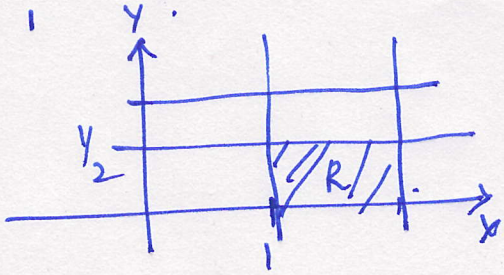


(iii) $P(X>1, Y<1/2)$

$= \iint_{x>1, y<1/2} (xy^2 + x^2/8)\,dx\,dy$

x lts 1 to 2
y lts 0 to 1/2

$= \int_0^{1/2} \int_1^2 (xy^2 + x^2/8)\,dx\,dy = 5/24$



iv. $P(X>1 / Y<1/2) = \dfrac{P(X>1, Y<1/2)}{P(Y<1/2)}$

$= \dfrac{5/24}{1/4} = 5/6$

v. $P(Y<1/2 / X>1) = \dfrac{P(Y<1/2, X>1)}{P(X>1)}$

$= \dfrac{5/24}{19/24} = \dfrac{5}{19}$.

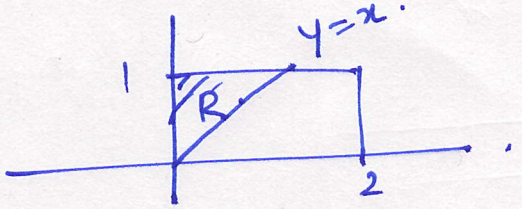
vi. $P(X<Y) = \iint_{X<Y} (xy^2 + x^2/8)\,dx\,dy$

Region of limits $0 \le y \le 1$ $0 \le x < y$

(since $y \le 1 \le 2$), the upper limit is $x = y$

$P(X<Y) = \int_{y=0}^{1} \int_{x=0}^{y} (xy^2 + x^2/8)\,dx\,dy = \dfrac{53}{480}$.

R.



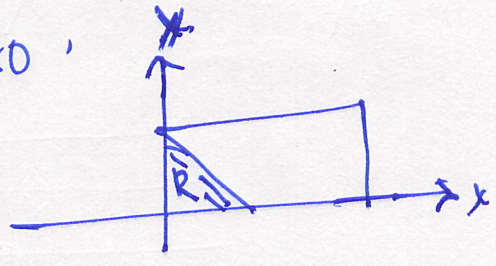
(vii) $P(X+Y \le 1) = \iint_{R=(x+y\le 1)} (xy^2 + x^2/8)\,dx\,dy$.

$0 \le y \le 1$

$0 \le x \le 1-y \le 2$.

$= \int_{y=0}^{1} \int_{x=0}^{1-y} (xy^2 + x^2/8)\,dx\,dy$

$= 13/480$.



Part A (Short answer questions)

1. Define a two-dimensional RV. Give an example for the outcome of a random experiment, that is a two-dimensional RV.
2. Define the joint pmf of a two-dimensional discrete RV.

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3. Define the joint pdf of a two-dimensional continuous RV.
4. Write down the joint pdf of a bivariate normal distribution.
5. Define the cdf of a two-dimensional RV and write down the formulas for, finding the cdf of (X, Y), when (X, Y) is (i) a discrete RV and (ii) a continuous RV.
6. State the properties of the cdf of a two-dimensional RV (X, Y).
7. Define the marginal probability distributions of X and Y, when (X, Y) is a discrete RV.
8. Define the marginal probability density functions of X and Y, when (X, Y) is a continuous RV.
9. Define independence of 2 RVs X and Y, both in the discrete case and in the continuous case.
10. Define the conditional probability distributions of X and Y, given Y and X respectively, when (X, Y) is a discrete RV.
11. Define the conditional probability density functions of X and Y given Y and X respectively, when (X, Y) is a continuous RV.
12. Find the probability distribution of $(X + Y)$ from the bivariate distribution of (X, Y) given below.

X	Y	
	1	2
1	0.1	0.2
2	0.3	0.4

13. Find the marginal distributions of X and Y from the bivariate distribution of (X, Y) given in Q.12.
14. Find the conditional distribution of X, when $Y = 1$, from the bivariate distribution of (X, Y) given in Q.12.
15. Find the value of k, if $f(x, y) = k(1 - x)(1 - y)$, for $0 < x, y < 1$, is to be a joint density function.
16. If $f(x, y) = k(1 - x - y)$, $0 < x, y < \frac{1}{2}$, is a joint density function, find k.
17. If the joint pdf of (X, Y) is $f(x, y) = \frac{1}{4}$, $0 \le x, y \le 2$, find $P(X + Y \le 1)$.
18. If the joint pdf of (X, Y) is $f(x, y) = 6e^{-2x-3y}$, $x \ge 0$, $y \ge 0$, find the marginal density of X and conditional density of Y given X.
19. The j pdf of (X, Y) is given by $f(x, y) = e^{-(x+y)}$, $0 \le x, y < \infty$. Are X and Y independent? Why?
20. Define a random vector with an example.
21. Define the joint density and distribution functions of an n-dimensional RV. How are they related?

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Part B

22. If X denotes the number of aces and Y the number of queens obtained when 2 cards are drawn at random (without replacement) from a deck of cards, obtain the joint probability distribution of (X, Y).
23. The joint probability function of two discrete RVs X and Y is given by $f(x, y) = c(2x + y)$, where x and y can assume all integers such that $0 \le x \le 2$ and $0 \le y \le 3$, and $f(x, y) = 0$ otherwise. (a) find the value of c and (b) find $P(X \ge 1, Y \le 2)$. (MKU — Apr. 96)

Note *$f(x, y)$ should not be mistaken as pdf, It is used instead of $p(x_i, y_i)$*

24. The joint probability distribution of a two-dimensional discrete RV (X, Y) is given below:

Y	X					
	0	1	2	3	4	5
0	0	0.01	0.03	0.05	0.07	0.09
1	0.01	0.02	0.04	0.05	0.06	0.08
2	0.01	0.03	0.05	0.05	0.05	0.06
3	0.01	0.02	0.04	0.06	0.06	0.05

(i) Find $P(X > Y)$ and $P\{\max(X, Y) = 3\}$ and
(ii) Find the probability distribution of the RV $Z = \min(X, Y)$.

25. The input to a binary communication system, denoted by a RV X, takes one of two values 0 or 1 with probabilities 3/4 and 1/4 respectively. Because of errors caused by noise in the system, the output Y differs from the input occasionally. The behaviour of the communication system is modeled by the conditional probabilities given below:
$$P(Y = 1/X = 1) = 3/4 \text{ and } P(Y = 0/X = 0) = 7/8$$
Find (i) $P(Y = 1)$, (ii) $P(Y = 0)$ and (iii) $P(X = 1/Y = 1)$.
26. The following table represents the joint probability distribution of the discrete RV (X, Y). Find all the marginal and conditional distributions.

Y	X		
	1	2	3
1	1/12	1/6	0
2	0	1/9	1/5
3	1/18	1/4	2/15

27. The joint distribution of X_1 and X_2 is given by $f(x_1, x_2) = \dfrac{x_1 + x_2}{21}$, $x_1 = 1, 2$ and 3; $x_2 = 1$ and 2. Find the marginal distributions of X_1 and X_2. (MU — Nov. 96).

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28. If the joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = x^2 + \frac{xy}{3}; \ 0 < x < 1, \ 0 < y < 2$$
$$= 0, \text{ elsewhere}$$

find (i) $P\left(X > \frac{1}{2}\right)$, (ii) $P(Y < X)$ and (iii) $P\left(Y < \frac{1}{2} / X < \frac{1}{2}\right)$.

29. If the joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = k(6 - x - y); \ 0 < x < 2, \ 2 < y < 4$$
$$= 0, \text{ elsewhere}$$

find (i) the value of k, (ii) $P(X < 1, Y < 3)$,
(iii) $P(X + Y < 3)$ and (iv) $P(X < 1/Y < 3)$.

30. The joint density function of the RVs X and Y is given by

$$f(x, y) = 8xy; \ 0 < x < 1, \ 0 < y < x$$
$$= 0, \text{ elsewhere,}$$

find $P\left(Y < \frac{1}{8} / X < \frac{1}{2}\right)$. (MU — Nov. 96)

31. Given that the joint pdf of (X, Y) is

$$f(x, y) = e^{-y}; \ x > 0, \ y > x$$
$$= 0, \text{ elsewhere}$$

find (i) $P(X > 1/Y < 5)$ and (ii) the marginal distributions of X and Y.

32. If the joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = 2; \ 0 < x < 1, \ 0 < y < x$$
$$= 0, \text{ otherwise}$$

find the marginal density function of X and Y. (BDU — Nov. 96)

33. If the joint pdf of (X, Y) is given by $f(x, y) = k$, $0 \le x < y \le 2$, find k and also the marginal and conditional density functions.

34. The joint density function of a RV (X, Y) is $f(x, y) = 8xy$, $0 < x < 1$, $0 < y < x$. find the conditional density function $f(y/x)$. (MU — Apr. 96)

35. The joint density function of a RV (X, Y) is given by $f(x, y) = axy$, $1 \le x \le 3$, $2 \le y \le 4$, and $= 0$, elsewhere.
Find (i) the value of a, (ii) the marginal densities of X and Y and (iii) the conditional densities of X and Y, given Y and X respectively.

36. Let X_1 and X_2 be two RVs with joint pdf given by $f(x_1, x_2) = e^{-(x_1 + x_2)}$; x_1, $x_2 \ge 0$, and $= 0$, otherwise. Find the marginal densities of X_1 and X_2. Are they independent? Also find $P[X_1 \le 1, X_2 \le 1]$ and $P(X_1 + X_2 \le 1)$.
(BDU — Apr. 96)

37. The joint pdf of the RVs X and Y is given by $p(x, y) = xe^{-x(y+1)}$ where $0 \le x, y < \infty$. (i) Find $p(x)$ and $p(y)$ and (ii) Are the RVs independent?
(BU — Nov. 96).

38. If the joint pdf of the RV (X, Y) is given by $f(x, y) = k(x^3y + xy^3)$, $0 \le x \le 2$, $0 \le y \le 2$, find (i) the value of k, (ii) the marginal densities of X and Y and (iii) the conditional densities of X and Y.